Title: Translating Open Science: The Turkish Adaptation of FORRT's Open Science Glossary

Short Title: Translating Open Science Glossary into Turkish

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Abstract

In response to the replication crisis, a significant number of structural and procedural changes have been made in how scientific research is conducted. Of these changes, open science practices, in particular, have been seen as a panacea for improving transparency, reproducibility, and accessibility in research. However, engaging in such reforms is not equally possible for everyone in academia, partly due to dominant resources being available in the English language. This situation results in a linguistic power asymmetry that creates and perpetuates structural barriers for researchers, educators, and students working in non-English contexts, overall limiting their efforts to take part in global scientific discourse. Using the frameworks of open science, linguistics, and epistemic justice as an organizing point, this article reports and reflects on translating the Open Science Glossary of the Framework for Open and Reproducible Research Training (FORRT) into Turkish, conducted as part of a multilingual initiative. Our team of five researchers carried out a collaborative translation process and translated approximately 250 open science terms. Translation decisions, particularly for terms without established Turkish equivalents, were debated collectively, and final decisions were made by considering the principles of clarity, accuracy, accessibility, and adaptation. We argue that translating terminology is an act of epistemic inclusion that supports capacity building, teaching, and research reform, rather than simply being a technical necessity. By localising open science concepts in the Turkish language, our work contributes to more inclusive and globally representative open science efforts.

Keywords: Open science, glossary, translation, replicability crisis, linguistic equity.

Introduction

Over the past decade, psychology and related disciplines have grappled with what has become known as *the replicability crisis*, in which many well-cited findings have proven difficult or impossible to reproduce (Camerer et al., 2018; Korbmacher et al., 2023; Open Science Collaboration, 2015; Tackett et al., 2019). This crisis highlighted systemic issues in research practices, including selective reporting, small sample sizes, and insufficient transparency. In response, the open science movement has emerged as a set of practices aimed at improving the credibility, replicability, and reproducibility of scientific research (Nosek et al., 2022; Wagenmakers et al., 2021).

Contemporary approaches to open science—also known as open research or open scholarship—encompass a range of practical tools and approaches that help researchers make their work more credible, robust, equitable, and transparent (Nosek et al., 2015). At the individual/researcher level, examples include preregistration of study protocols (e.g., through Open Science Framework) to reduce bias, sharing data and materials openly to allow independent verification and reuse, and registered reports, in which study protocols are peer-reviewed and accepted in principle before data collection to minimize publication bias and questionable research practices. Beyond individual practices, open science also involves community-based and infrastructural initiatives. Community-based efforts include the democratization of open science support and knowledge through community (Azevedo et al., 2019), participatory (Gourdon-Kanhukamwe et al., 2023), and big-team science approaches (Korbmacher et al., 2023), as well as replication initiatives (Röseler et al., 2024) and collaborative researcher networks (Skubera et al., 2025). Infrastructural initiatives provide the structural backbone for these efforts, including databases of open science resources (Azevedo et al., 2019), dedicated handbooks on reproduction and replication research, and dedicated diamond open access journals (Röseler et al., 2024). One such initiative that integrates these diverse efforts is the Framework for Open and Reproducible Research Training (FORRT), which serves as a coordinating epistemic scaffold operating primarily at the infrastructural and pedagogical level to support and integrate open scholarship practices. Taken together, these practices aim to make research more generalizable, improve the system of incentives, and facilitate uptake, ultimately fostering the cumulative building of knowledge (Hales et al., 2019; Munafò et al., 2017).

Despite these advances, participation in open science reforms is not equally distributed across global academic communities. Open science resources, frameworks, and terminologies have largely been developed within Western academic contexts and are predominantly disseminated in English language, a dynamic that can unintentionally reproduce *linguistic and epistemic barriers* to equitable participation in scientific knowledge production. As a result, scholars, students, and educators working in non-English contexts may face additional challenges in accessing, interpreting, and meaningfully engaging with open science concepts. Addressing these inequalities requires not only expanding access but also recognizing translation as a process that reshapes how knowledge itself is constituted across linguistic contexts, rather than a secondary act of dissemination (Spivak, 2021; Venuti, 1995).

Linguistic and Epistemic Equity in Open Science

Linguistic barriers shape who can meaningfully participate in open science reforms and whose methodological knowledge becomes visible. Although open science aims to democratize research practices, much of the discourse, resources, and terminology associated with open science remain concentrated in English. As a result, access to open science is uneven. Non-English-speaking academic communities, especially those from the Global South, often face difficulties while engaging with methodological reforms and integrating open science principles into teaching, supervision, and everyday research practice. Linguistic exclusion therefore constitutes a form of structural inequality within global science: scholars and students with limited English proficiency experience higher cognitive load, slower training trajectories, restricted access to methodological debates, and reduced publication opportunities (Amano et al., 2016, 2023).

From this perspective, translation becomes central to addressing inequities in open science participation. Translation is not merely a linguistic task, but it is closely tied to epistemic participation. When methodological terminology exists only in English, entire academic communities become structurally dependent on Anglophone gatekeepers. Such dependence reflects broader patterns of epistemic injustice, in which access to knowledge production and shared interpretive resources is unequally distributed (Fricker, 2007), as well as structural inequities embedded in academic publishing (Curry & Lillis, 2004). Providing terminology in additional languages can help address these asymmetries by facilitating access to key methodological concepts, reducing linguistic marginalization, and supporting broader participation in open science discourse. In this sense, we see translation as an act of epistemic empowerment.

Conceptually, we understand epistemic justice as concerning fairness in knowledge practices, specifically, who is able to contribute to knowledge production, whose contributions are recognised and valued, and who has access to shared interpretive resources (Fricker, 2007). Linguistic justice, on the other hand, concerns the equitable distribution and recognition of linguistic resources within institutional structures, including which languages are treated as legitimate vehicles of scholarly communication (Curry & Lillis, 2004; Van Parijs, 2011). In the context of global open science, we see linguistic justice as a structural precondition of achieving epistemic justice. Without equitable access to methodological language, meaningful epistemic engagement will remain constrained. By enabling Turkish-speaking researchers and students to understand and teach open science concepts in their own language, our translated glossary supports more inclusive knowledge infrastructures and contributes to broadening the epistemic diversity of traditionally English-dominant scientific ecosystems.

Multilingual Open Science Initiative: The Turkish Translation of the FORRT Glossary

One international initiative addressing the abovementioned barriers is the Framework for Open and Reproducible Research Training (FORRT), which has developed an open science glossary to clarify key terms, foster common understanding, and serve as a pedagogical resource (Azevedo et al., 2019; *Glossary*, 2021; Parsons et al., 2022). By translating this glossary into multiple languages, FORRT aims to promote inclusivity and ensure that the benefits of open science extend beyond English-speaking contexts. From the perspective of epistemic justice (Fricker, 2007), providing key methodological terminology in Turkish enhances equitable access to scientific knowledge by reducing the exclusionary effects of linguistic barriers and increasing the availability of open science related content.

Within the Turkish context, several initiatives have sought to address this gap. Community-driven platforms such as Açık Bilim[1] Topluluğu Türkiye contributed educational materials and reflective discussions on open science, supporting grassroots engagement with open research practices (*Açık Bilim Topluluğu Türkiye*, n.d.). At the institutional level, APERTA (*Aperta*, n.d.), provided by the Scientific and Technological Research Council of Türkiye (TÜBİTAK), represents an important policy-oriented effort to support open access infrastructures, such as preregistration. In addition, public-facing science communication websites, such as AcikBilim.com, play a role in promoting scientific literacy in Turkish, albeit without a specific focus on systematically defining or standardizing open science terminology (*Açık Bilim*, n.d.).

[1] *Açık Bilim* is the Turkish translation commonly used for *Open Science* and is increasingly adopted across academic, institutional, and community-based initiatives in Türkiye.

Taken together, these resources have made valuable contributions to open science awareness in Türkiye. However, they vary in scope, conceptual depth, and terminological consistency, and none currently offers a comprehensive, curated glossary aligned with international open science discourse. In response to these gaps, our Turkish translation of FORRT's Open Science Glossary is intended as a complementary and necessary tool that seeks to reduce linguistic barriers, enhance conceptual clarity, and promote more inclusive engagement with open science. We hope this resource will serve as an essential reference for undergraduates, postgraduate students, early-career researchers, student–academics, educators, and all individuals interested in developing a deeper and more precise understanding of open science principles and practices in Turkish.

In this context, our team undertook the task of translating the FORRT's open science version 1.0 glossary into Turkish. Our aim was to make the principles and practices of open science—such as preregistration, open data, and registered reports—accessible to Turkish-speaking students, researchers, and educators, thereby supporting more transparent and replicable research in psychology and related fields. Beyond producing a reference tool, the translated glossary also has pedagogical implications. Providing key methodological terminology in Turkish allows instructors to teach open science concepts in their native language, lowers barriers for first-generation students, and supports early-career researchers who may have limited English proficiency. For many students—particularly those entering academia from under-resourced or non-elite educational backgrounds—the availability of Turkish terminology may be their first point of access to open science concepts. The glossary therefore supports equity not only across languages but also across socioeconomic strata. In doing so, it broadens access to methodological reform and strengthens the educational mission at the core of open science.

In addition, the translation work carried out by the Turkish team was part of a broader multilingual effort coordinated by FORRT, in which more than ten language groups collaborated to expand the glossary from roughly 250 to over 800 terms. Situating the Turkish contribution within this international initiative underscores how multilingual collaboration fosters shared methodological standards, supports capacity building across communities, and reveals cross-cultural nuances in the conceptualization of open science.

In the sections that follow, we describe our collaborative translation process, the challenges we encountered, and the solutions we developed. We also discuss how the translated Turkish glossary—now available on FORRT's website (*Sözlük*, 2026)—can be used to promote open science education and advance replicability efforts within Turkish academia.

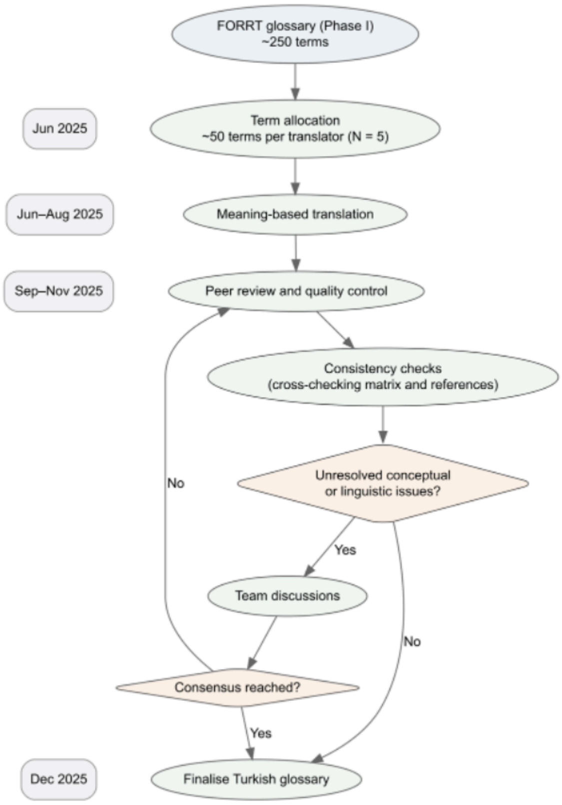
Methods

Translation Process

The Turkish translation of the FORRT's open science glossary was carried out by a team of five researchers with backgrounds in various subfields of psychology. The translation process was designed to ensure accuracy, consistency, and accessibility, while fostering collaboration and collective decision-making. Rather than adopting a strictly verbatim (word-by-word) approach, translators prioritized meaning-based translation, aiming to convey the intended conceptual content of each term in a way that is linguistically natural and contextually meaningful in Turkish. The translation process took approximately five months, from June 2025 to November 2025. Figure 1 illustrates the translation process. See Table S1 within the Supplemental Material for a table of example translations.

Figure 1

Workflow of the Translation Process for the Turkish Version of the FORRT Glossary



Workflow and Collaboration

The FORRT's open science glossary (version 1.0) currently contains approximately 250 key terms related to open science. In the current English glossary, each term is presented with its brief definition, alongside related terms and references. To start, these terms were evenly distributed among the five researchers, with each team member initially responsible for translating around 50 terms. This division allowed for efficient workflow while ensuring that each term received focused attention.

Translation proceeded iteratively over several months. The team held regular monthly meetings to discuss progress, share challenges, and align on translation principles. Translators were encouraged to flag terms when needed: for instance, when a direct Turkish equivalent did not exist or conceptual clarity was difficult to preserve. These terms were subsequently discussed within the team to reach a consensus. During this process, the team focused intensely on value-laden translation choices (e.g., anglicisms vs. localized neologisms), acknowledging that these decisions participate in shaping the normative vocabulary of open science in Turkish. In line with open science and transparency principles, we note that artificial intelligence–based tools were consulted as supportive instruments during the translation process, mainly to explore alternative expressions and to improve clarity.

To facilitate collaborative review, tags such as *review needed* were used to prompt not only the designated reviewer but all team members to review specific terms that were difficult to translate. To ensure internal consistency, the team implemented a cross-checking matrix tracking recurrence of terms across definitions. This allowed us to identify definitional dependencies and harmonize terminology at the glossary-wide level. Color coding[2] was also used for highlighting terms that appeared on other translators' term lists or recurred in definitions, ensuring these terms were checked for consistency. Terms that were difficult or problematic to translate were highlighted in a different color.

[2] For instance, terms were highlighted in red if the term within the definition was in another team member's translation list; and in pink if the team member was not sure about the translation of the term and wanted at least one more team member to revise.

Quality Control

Each translated term was reviewed by at least one other team member to ensure accuracy, readability, and consistency. In cases of disagreement or ambiguity, the term was reviewed by the entire team during regular meetings, and a collective decision was made on the best translation. This approach allowed us to balance faithfulness to the original English meaning with linguistic and cultural appropriateness for Turkish-speaking readers.

Decision Principles

Throughout the process, the team adhered to key guiding principles. One of the guiding principles was *clarity.* For this, translations had to be understandable to under- and post-graduate students, early-career researchers, and educators. Second, *accuracy* was sought for translations. For this, the team aimed the terms to reflect the intended methodological or conceptual meaning. Next, the team had a principle for *consistency.* For this, the team worked carefully on terms to maintain uniformity with previously translated entries across the glossary. Lastly, *accessibility* was a key principle. Translations had to avoid overly technical language or unnecessary anglicisms, where possible, to increase readability.

Additionally, during the translation process, *adaptation* was prioritised over verbatim translation provided in the original text to ensure contextual and temporal appropriateness. For example, with regards to direct replication, the original text stated: *''A technical equivalent replication would use Trump as the stimulus (he was president in 2018), whereas a psychologically equivalent study would use Biden (he is the current president).''* However, because Biden was no longer the president of the United States at the time of translation, a direct translation would have conveyed outdated information. Therefore, to preserve the intended meaning in the original text and ensure temporal accuracy, this information was adapted in the Turkish version, with additional contextual clarification provided.

Challenges and Resolutions

During the translation process, the team encountered several challenges that required careful deliberation and collaborative problem-solving. These challenges primarily revolved around terminological equivalence, clarity, and consistency. Final decisions were reached collectively, often with input from multiple team members to ensure both conceptual fidelity and accessibility for Turkish-speaking readers with diverse levels of methodological expertise.

Lack of Direct Turkish Equivalents

Some open science terms had no direct or widely accepted Turkish equivalents (i.e., conceptually challenging terms) (e.g., *preregistration*, *registered report*). In several cases, the absence of Turkish equivalents revealed how deeply methodological vocabulary is rooted in Anglophone epistemic cultures. This prompted discussions about whether to create new terms, adapt existing Turkish methodological terminology, or retain English expressions. These negotiations highlight how translation participates in constructing the conceptual architecture of local scientific practice. In such cases, the team followed a structured decision-making process built around three primary translation principles. The selection of the most appropriate rendering for technically dense terms was guided by these three principles, which were applied contextually on a case-by-case basis rather than hierarchically in a fixed or preferential order.

- Transliteration, preserving phonetic proximity to the English term (e.g., *pseudoreplication → psödoreplikasyon*).
- Literal or compositional translation, combining Turkish morphemes to reflect the structure of the original term (e.g., *preregistration → ön-kayıt; paywall → ödeme duvarı*).
- Descriptive translation, using short explanatory phrases that captured the operational meaning of the concept (e.g., *overlay journal → ön-baskı tabanlı dergi).*

Balancing Technical Accuracy and Accessibility

A further challenge involved terms that are highly technical and well established within methodological and statistical literature. The challenge was to preserve precise methodological meaning while keeping the language readable for students and early-career researchers. During our translation process, we frequently encountered a dilemma between linguistic naturalness and methodological precision. For instance, while a fully localized translation of *multiverse analysis* would read more fluently in Turkish, such phrasing risked losing conceptual specificity.

– Recognizability within the international literature, favoring retention of English terms when they function as stable methodological labels (e.g., *HARKing → HARKing, Sonuçlar Bilindikten Sonra Hipotez Belirleme [Hypothesizing After the Results are Known])*.
– Learnability for early-career researchers, prioritizing formulations that reduce cognitive load without oversimplifying the concept (e.g., *salami slicing → veri dilimleme*).
– Linguistic and cultural fit, ensuring that adopted forms conformed to Turkish morphological and pragmatic norms (e.g., *p-hacking → p-hackleme*, *Red Teams → kızıl takımlar).*

Maintaining Consistency Across the Glossary

With multiple translators working in parallel, consistency in terminology and style was critical. To support this process, we created a shared reference document with finalized translations of recurring terms, which all translators consulted before finalizing new translations. In addition, terms translated by another translator and terms recurring in other translations were systematically checked during the review process to ensure consistency. In parallel, each translator reviewed their own assigned term list recurrently to ensure within-translator consistency in style and terminology. Regular monthly team meetings also provided an opportunity to align on stylistic conventions, phrasing, and formatting decisions.

Resolving Disagreements

Disagreements were inevitable, especially for nuanced conceptual terms. The team resolved these through discussion and consensus, guided by the principles of clarity, accuracy, and accessibility. For particularly challenging terms, the team sometimes consulted external references or examples from existing Turkish methodological literature.

In addition to these considerations, the team adhered, where appropriate, to terminology consistent with the conventions of the Turkish Language Association (Türk Dil Kurumu; TDK), particularly for well-established methodological constructs. When multiple Turkish renderings were in circulation, preference was given to forms aligned with TDK usage in order to promote terminological consistency and institutional recognizability (e.g., *construct validity* → *yapı geçerliliği* rather than *yapı geçerliği*). This practice functioned as a stabilizing constraint within the translation process, helping to balance innovation in emerging open science terminology with continuity in established methodological language.

Reflexivity in Translation and Power Dynamics

Translation is not a neutral transfer of meaning but a situated practice shaped by global hierarchies of knowledge production. Throughout our work, we observed how English-language methodological discourse functioned as a dominant epistemic framework, subtly influencing which formulations appeared *standard, legitimate* or *authoritative*. These dynamics highlighted the power relations embedded in translation decisions, from choosing between anglicisms and localized neologisms to determining which epistemic traditions would be privileged or marginalized.

Recognising that such choices carry implicit power relations, we approached translation with deliberate reflexivity, continually asking whose linguistic and methodological traditions were being centered or sidelined. This reflexive stance is essential for multilingual science reform efforts seeking to disrupt rather than reproduce existing epistemic asymmetries.

Reflexivity also informed the way we handled disagreements. Rather than deferring to perceived expertise, the team engaged in transparent, non-hierarchical deliberation, aligning the translation process with FORRT's broader commitment to distributed leadership and community-driven knowledge creation.

In addition, while we used artificial intelligence–based tools, the translation process was not automated. The authors served as the primary translators and researchers, making all conceptual, linguistic, and methodological decisions and assuming full responsibility for the final content.

Ultimately, approaching translation reflexively allowed us to treat it as an epistemic practice rather than a mechanical substitution. The process illuminated how linguistic choices help construct the conceptual landscape of open science in Turkish and shape how methodological ideas are adopted, taught, and institutionalized across academic contexts.

Outcome

By systematically addressing these challenges, the team produced a glossary that is conceptually faithful to the original English version, linguistically accessible for Turkish readers, and internally consistent across all 250 terms. The collaborative approach ensured transparency and quality control, providing a replicable model for future translation efforts in academic contexts.

Discussion

We translated FORRT's open science glossary version 1.0 into Turkish using a systematic, principle-based, collaborative, and participatory methodology. Decisions were made collectively, disagreements resolved transparently, and no single translator held hierarchical authority, ensuring accurate, consistent, and contextually appropriate terminology.

Significance and Implications

The translation of the FORRT open science glossary into Turkish represents more than a linguistic exercise: it is a capacity-building effort aimed at lowering barriers to open science engagement for Turkish-speaking communities. By making foundational concepts available in Turkish, the glossary provides an entry point for students, early-career researchers, and educators to adopt practices that enhance transparency, replicability, and reproducibility. This is particularly important in the context of psychology and related fields, where the replicability crisis has underscored the need for methodological reform and stronger research practices (Francis, 2012; Wicherts et al., 2016). Our work also aligns with recent scholarship on multilingual open science, which emphasizes that linguistic equity is critical for democratizing participation in research reform (Ramírez-Castañeda, 2020). By making methodological terminology accessible to Turkish-speaking scholars, the glossary contributes to a more inclusive infrastructure for transparent scientific practice.

Beyond its immediate value as a reference tool, the glossary has significant pedagogical potential. It can be incorporated into research methods and statistics courses, serve as a resource for thesis and dissertation supervision, and support professional training workshops. To maximize accessibility, the glossary is freely available on the FORRT website along with English, German, and Arabic versions (*Sözlük*, 2026). Our team also plans to publish the

Turkish glossary as an open access booklet with a DOI, ensuring it can be easily cited and disseminated across academic networks.

The contribution of this project should also be considered within the broader structural context of open science practices in Türkiye. Although recent top-down initiatives by national bodies such as major funding agencies (e.g., TÜBİTAK) have led to the development of national open access platforms, repository systems, and formal open science policies, these efforts have not yet translated into widespread or routinized research practices at the researcher level. Prior research indicates that awareness and engagement with open science and open research data remain limited, and that many initiatives continue to rely heavily on voluntary and uncompensated academic labor rather than sustained institutional support across different disciplinary contexts (Ataman et al., 2021; Zencir & Oğuz, 2020).

The significance of our translation effort can also be understood through the broader theoretical lens of linguistic equity and knowledge localization. Scholarship on multilingual science argues that when key methodological terms remain available exclusively in English, researchers in non-English-speaking communities face epistemic disadvantages that hinder their participation in scientific discourse (Meneghini & Packer, 2007). Parallel work in academic writing research highlights how English-language dominance shapes whose knowledge becomes visible and valued, reinforcing structural inequalities in global science (Curry & Lillis, 2004; Tardy, 2004). By providing Turkish-language terminology for foundational open science concepts, our glossary contributes to reducing these inequities and supports the localization of methodological knowledge—a process shown to facilitate wider adoption of scientific reforms across diverse academic communities (Ramírez-Castañeda, 2020). This perspective positions translation not merely as linguistic transfer but as an act of epistemic inclusion within the open science landscape.

From a theory of change (ToC) perspective, translation can be conceptualized not as a passive act of access facilitation but as a deliberate upstream intervention designed to create the enabling conditions necessary for methodological reform. A ToC framework makes explicit the causal pathways, intermediate conditions, and assumptions linking activities to long-term impact (Anderson & Change, 2006; Mayne, 2017; Reinholz & Andrews, 2020). Consistent with the Center for Open Science's Pyramid of Change (Nosek, 2019), which emphasizes shared understanding and infrastructure in shifting research norms, translation strengthens conceptual accessibility and shared language at an early stage of the change process. In this context, providing open science terminology in Turkish is not an end in itself but an intervention intended to improve the cognitive and pedagogical accessibility of open science practices, thereby helping establish the foundations for later behavioral and institutional shifts.

In line with core ToC principles, complex social outcomes emerge through causal pathways that involve multiple intermediate conditions rather than immediate behavioral effects (Anderson & Change, 2006; Mayne, 2017). Within this pathway, the long-term outcome—greater adoption of open science in Turkish academic contexts—depends on several intermediate conditions. Methodological concepts must first become cognitively accessible, reducing the processing demands associated with specialized terminology in a second language. Shared terminology then enables deeper classroom interaction, supervision, and peer discussion, shifting learning from surface-level comprehension to conceptual elaboration. These shifts are expected to increase perceived competence, self-efficacy, and practical feasibility among researchers, thereby strengthening intentions to engage in open science practices. Ultimately, when the underlying assumptions hold—that reduced linguistic barriers facilitate learning and that conceptual clarity enhances perceived implementability—the Turkish glossary functions as a structural mechanism that makes methodological reform learnable, teachable, and actionable. Rather than merely representing symbolic inclusion, it serves as an enabling resource for the institutional change processes required for downstream behavioral adoption (Mayne, 2017; Reinholz & Andrews, 2020).

Limitations and Future Work

While translating, no minimal criteria (e.g., a predefined set of databases or sources to be systematically checked) were established when searching for whether existing translations already existed. Consequently, multiple translations of certain terms may have emerged. Such variability is not inherently problematic, as different translations can be appropriate in different contexts; however, it may pose challenges for consistency across studies or applications. For future translation work, establishing minimal criteria may help improve consistency. Additionally, we were unable to receive feedback from pilot testers (e.g., psychology students) and conduct comprehension checks with independent observers (e.g., language experts). Future work may include updating the glossary as open science practices evolve, integrating it into course curricula, and promoting it through workshops, webinars, and professional societies. One direction could include combining glossary translation with educational practice. Similar to how replication is incorporated as an educational activity within the Collaborative Replication and Education Project (CREP)[3], translating glossaries into other languages by students could likewise be developed into a structured educational activity (Wagge et al., 2019). Importantly, embedding glossary translation within educational settings may also provide a sustainable and scalable model for translating open science terms, reducing reliance on a small number of researchers.

[3] Collaborative Replication and Education Project (CREP) is an initiative in which students and researchers collaborate to conduct high-quality direct replications of influential studies, addressing the dual needs for more reliable replication evidence and stronger training in empirical research methods.

Beside limitations, the translation process revealed opportunities for coordination across other FORRT translation teams—for example, developing a shared multilingual style guide, co-authoring a meta-paper examining cross-linguistic challenges, or collectively identifying concepts that do not translate cleanly across linguistic families. Pursuing such collaborations could strengthen the global coherence of multilingual open science infrastructure and increase both scholarly impact and publication potential. More broadly, we hope that our participatory translation model can inspire similar efforts in other linguistic and disciplinary contexts, contributing to the inclusivity and global reach of open science.

Conclusions

The replicability crisis has made clear that psychology and related fields must prioritize transparency and reproducibility to strengthen scientific knowledge. Open science practices offer concrete solutions to these challenges, but their adoption is often hindered by linguistic barriers. By translating the FORRT open science glossary into Turkish through a collaborative, principle-based process, we aimed to bridge this gap and provide the Turkish-speaking academic community with accessible, accurate terminology.

This glossary serves both as a reference tool and as a teaching resource, supporting the integration of open science practices into research and education. Making open science concepts available in the Turkish language is a step toward democratizing access to methodological reform, fostering replicable research, and building stronger connections between Turkish-speaking academia and the global scientific community.

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Ethics statement

This study did not involve testing of human participants. The translation was carried out in line with the ethical principles of the FIT Translator's Charter, with careful attention to accuracy and fidelity to the original source.

Competing interests

The authors have no conflicts of interest to declare.

Supplemental material

Table S1. Example translations from open science glossary version 1.0.

Data accessibility statement

Not applicable.

Figure titles and legends

Figure 1. Workflow of the Translation Process for the Turkish Version of the FORRT Glossary.

Supplemental Material

Table S1. Example translations from open science glossary version 1.0.

Term	**Definition**	**Turkish Term**	**Turkish Definition**
Salami slicing	A questionable research/reporting practice strategy, often done *post hoc*, to increase the number of publishable manuscripts by 'slicing' up the data from a single study - one example of a method of 'gaming the system' of academic incentives. For instance, this may involve publishing multiple studies based on a single dataset or publishing multiple studies from different data collection sites without transparently stating where the data originally derives from. Such practices distort the literature, and particularly meta-analyses, because	**Veri Dilimleme**	Şüpheli bir araştırma/raporlama stratejisi, genellikle *post hoc* gerçekleştirilir ve tek bir çalışmadan elde edilen veriler 'dilimlenerek' birden fazla yayımlanabilir makale üretilmesi amaçlanır. Bu, akademik teşvik sistemini manipüle etmenin bir yoludur. Örneğin, tek bir veri setinden birden fazla çalışma yayımlamak ya da farklı veri toplama merkezlerinden gelen verilerle yapılmış çalışmaları, verinin kaynağını şeffaf biçimde belirtmeden yayımlamak bu kapsamdadır. Bu tür uygulamalar, alan yazını ve özellikle meta-

	it is unclear that the findings were obtained from the same dataset, thereby concealing the dependencies across the separately published papers.		analizleri yanıltır, çünkü elde edilen bulguların aslında aynı veri setinden kaynaklandığı anlaşılmaz, böylece birbirinden ayrı yayımlanmış makaleler arasındaki bağımlılıklar gizlenmiş olur.
Scooping	The act of reporting or publishing a novel finding prior to another researcher/team. Survey-based research indicates that fear of being scooped is an important fear-related barrier for data sharing in psychology, and agent-based models suggest that competition for priority harms scientific reliability (Tiokhin et al. 2021).	**Önceliği Kapma**	Bir araştırmacı veya araştırma ekibinin, başka bir araştırmacıdan/ekipten önce özgün bir bulguyu raporlaması veya yayımlaması eylemidir. Psikoloji alanında yapılan anket bazlı araştırmalar, önceliği kaptırma korkusunun veri paylaşımının önündeki korkuya dayalı önemli engellerden biri olduğunu göstermektedir. Ajan tabanlı modellemeler (agent-based models) ise, öncelik için rekabetin bilimsel güvenilirliği olumsuz etkilediğini öne sürmektedir (Tiokhin vd., 2021).

Slow science	Adopting Open Scholarship practices leads to a longer research process overall, with more focus on transparency, reproducibility, replicability and quality, over the quantity of outputs. Slow Science opposes publish-or-perish culture and describes an academic system that allows time and resources to produce fewer higher-quality and transparent outputs, for instance prioritising researcher time towards collecting more data, more time to read the literature, think about how their findings fit the literature and documenting and sharing research materials instead of running additional studies.	**Yavaş Bilim**	Açık Akademi uygulamalarının benimsenmesi, araştırma sürecini genel olarak uzatır; çünkü bu süreçte şeffaflık**,** yeniden üretilebilirlik, tekrarlanabilirlik ve nitelik, çıktı sayısından daha çok önemsenir. Yavaş bilim, "yayımla ya da yok ol" (publish-or-perish) kültürüne karşı çıkar ve daha az ama daha kaliteli ve şeffaf çıktıların üretilebildiği, zamana ve kaynaklara olanak tanıyan bir akademik sistemi tanımlar. Örneğin, bu yaklaşımda araştırmacıların zamanları; daha fazla veri toplamak, alan yazını daha derinlemesine okumak, bulgularının alan yazınla nasıl örtüştüğünü düşünmek, araştırma materyallerini belgelemek ve paylaşmak gibi nitelikli süreçlere yönlendirilir. Böylece, yeni çalışmalar yapma

			baskısı yerine araştırmanın sağlamlığını artırmaya odaklanılır.
Credibility revolution	The problems and the solutions resulting from a growing distrust in scientific findings, following concerns about the credibility of scientific claims (e.g., low replicability). The term has been proposed as a more positive alternative to the term replicability crisis and includes the many solutions to improve the credibility of research, such as preregistration, transparency, and replication.	**İtibar Devrimi**	Bilimsel bulgulara yönelik artan güvensizlikle birlikte ortaya çıkan sorunları ve bu sorunlara yönelik çözümleri ifade eder. Bu güvensizlik, özellikle bilimsel iddiaların itibarına ilişkin endişelerden (örneğin, düşük tekrarlanabilirlik) kaynaklanmaktadır. Bu terim, tekrarlanabilirlik krizi terimine daha olumlu bir alternatif olarak önerilmiş olup, ön-kayıt, şeffaflık ve replikasyon çalışmaları gibi bilimsel araştırmaların itibarını artırmaya yönelik çeşitli çözüm yollarını kapsar.
Epistemic uncertainty	Systematic uncertainty due to limited data, measurement precision, model or process	**Epistemik Belirsizlik**	Sınırlı veri, ölçüm hassasiyetinin zayıflığı, model ya da süreç tanımlamalarındaki eksiklikler ya da

	specification, or lack of knowledge. That is, uncertainty due to lack of knowledge that could, in theory, be reduced through conducting additional research to increase understanding. Such uncertainty is said to be personal, since knowledge differs across scientists, and temporary since it can change as new data become available.		bilgi eksikliğinden kaynaklanan sistematik bir belirsizliktir. Yani, bilgi eksikliğinden doğan ve teoride, daha fazla araştırma yapılarak ve konunun daha iyi anlaşılmasıyla azaltılabilecek bir belirsizliktir. Bu tür belirsizlik kişiseldir, çünkü bilgi düzeyi bilim insanları farklılık gösterebilir; aynı zamanda geçicidir, çünkü yeni veriler elde edildikçe bu belirsizlik durumu değişebilir.
Garden of forking paths	The typically-invisible decision tree traversed during operationalization and statistical analysis given that 'there is a one-to-many mapping from scientific to statistical hypotheses' (Gelman and Loken, 2013, p. 6). In other words, even in absence of p-hacking or fishing expeditions and when the research hypothesis was posited ahead of time,	**Yolları Çatallanan Bahçe**	Yolları çatallanan bahçe, operasyonelleştirmeden istatiksel analize kadar uzanan araştırma sürecinde genellikle görünmeyen bir karar ağacının izlenmesini ifade eder; çünkü "bilimsel bir hipotezin istatistiksel olarak test edilmesinin birden fazla yolu vardır" (Gelman ve Loken, 2013, s. 6). Başka bir deyişle, araştırma hipotezi önceden

	there can be a plethora of statistical results that can appear to be supported by theory given data. “The problem is there can be a large number of potential comparisons when the details of data analysis are highly contingent on data, without the researcher having to perform any conscious procedure of fishing or examining multiple p-values” (Gelman and Loken, 2013, p. 1). The term aims to highlight the uncertainty ensuing from idiosyncratic analytical and statistical choices in mapping theory-to-test, and contrasting intentional (and unethical) questionable research practices (e.g. p-hacking and fishing expeditions) versus non-intentional research practices that can, potentially, have the same effect despite not having intent to corrupt their results.		belirlenmiş ve p-hacking (p-değeriyle oynama) ya da bilinçli veri avcılığı gibi şüpheli uygulamalara başvurulmamış olsa bile, eldeki verilerle teoriyi destekliyor gibi görünen pek çok istatistiksel sonuç üretmek mümkündür. “Sorun şu ki: analiz sürecinin detayları veriye bağlı olarak şekillendiğinde, araştırmacı bilinçli olarak veri avcılığı yapmasa ya da çok sayıda p-değerini incelemese bile, olası karşılaştırmaların sayısı oldukça fazla olabilir” (Gelman ve Loken, 2013, s. 1). Bu terim, teoriden teste geçişte yapılan öznel analitik ve istatistiksel tercihlerden kaynaklanan belirsizliklere dikkat çekmeyi amaçlar. Bununla birlikte, bilinçli (ve etik dışı) şüpheli araştırma uygulamalarıyla (örn., p-hacking veya veri

	The garden of forking paths refers to the decisions during the scientific process that inflate the false-positive rate as a consequence of the potential paths which could have been taken (had other decisions been made).		avcılığı) bilinçli olmadan yapılan ancak benzer şekilde yanıltıcı sonuçlara yol açabilecek uygulamaları karşılaştırır. Yolları çatallanan bahçe, bilimsel süreçte alınan kararların, alternatif yolların (başka kararlar alınmış olsaydı izlenebilecek yolların) varlığı nedeniyle yanlış pozitif oranını artırması durumunu ifade eder.
Conceptual replication	A replication attempts whereby the primary effect of interest is the same but tested in a different sample and captured in a different way to that originally reported (i.e., using different operationalisations, data processing and statistical approaches and/or different constructs; LeBel vd., 2018). The purpose of a conceptual replication is often to explore what conditions limit the extent to	**Kavramsal Replikasyon**	İlgilenilen birincil etkinin aynı kaldığı, ancak bu etkinin farklı bir örneklem üzerinde ve orijinal çalışmadan farklı yollarla test edildiği tekrar çalışmasıdır (örneğin, farklı işlemselleştirmeler, veri işleme ve istatistiksel yaklaşımlar ve/veya farklı yapılar kullanılarak; bkz. LeBel vd., 2018). Kavramsal tekrarlamanın amacı genellikle, bir etkinin gözlemlenebilirlik ve genelleştirilebilirlik

	which an effect can be observed and generalised (e.g., only within certain contexts, with certain samples, using certain measurement approaches) towards evaluating and advancing theory (Hüffmeier et al., 2016).		derecesini (örn., yalnızca belirli bağlamlarda belirli örneklemelerle, belirli ölçüm yaklaşımları kullanılarak) hangi koşulların sınırladığını araştırarak teoriyi değerlendirmek ve ilerletmektir (Hüffmeier vd., 2016).
Bropenscience	A tongue-in-cheek expression intended to raise awareness of the lack of diverse voices in open science (Bahlai, Bartlett, Burgio et al. 2019; Onie, 2020), in addition to the presence of behavior and communication styles that can be toxic or exclusionary. Importantly, not all bros are men; rather, they are individuals who demonstrate rigid thinking, lack self-awareness, and tend towards hostility, unkindness, and exclusion (Pownall et al., 2021; Whitaker & Guest, 2020). They generally	**Agaçık Bilim**	Açık bilimde çeşitli seslerin eksikliğine (Bahlai, Bartlett, Burgio vd., 2019; Onie, 2020) ve toksik veya dışlayıcı olabilen davranış ve iletişim tarzlarına dikkat çekmeyi amaçlayan esprili bir ifade. Önemli olan, tüm agaların ("bro"ların) erkek olmamasıdır; aksine, bunlar katı düşünce tarzı sergileyen, öz farkındalıktan yoksun, düşmanlık, kaba davranış ve dışlayıcı eğilimleri olan bireylerdir (Pownall vd., 2021; Whitaker & Guest, 2020). Genellikle yapısal ayrıcalıklardan

	belong to dominant groups who benefit from structural privileges. To address #bropenscience, researchers should examine and address structural inequalities within academic systems and institutions.		yararlanan egemen gruplara mensupturlar. #agaçıkbilim (#bropenscience) sorununu ele almak için, araştırmacıların akademik sistemler ve kurumlar içerisindeki yapısal eşitsizlikleri sorgulaması ve çözmeye çalışması gerekmektedir.
Open washing	Open washing, termed after "greenwashing", refers to the act of claiming openness to secure perceptions of rigor or prestige associated with open practices. It has been used to characterise the marketing strategy of software companies that have the appearance of open-source and open-licensing, while engaging in proprietary practices. Open washing is a growing concern for those adopting open science practices as their actions are undermined by misleading uses of the practices,	**Açık Aklama**	'Yeşil aklama (greenwashing)' teriminden türetilen açık aklama (open washing), açıklık uygulamalarıyla ilişkilendirilen titizlik veya prestij algısını sağlamak amacıyla açıklık iddiasında bulunma eylemini ifade eder. Bu terim, açık kaynaklı ve açık lisanslı gibi görünen ancak aslında tescilli uygulamalarda bulunan yazılım şirketlerinin pazarlama stratejisini tanımlamak için kullanılmıştır. Açık aklama, açık bilim uygulamalarını benimseyenler için giderek artan

	and actions designed to facilitate progressive developments are reduced to ‘ticking the box’ without clear quality control.		bir endişe kaynağıdır; çünkü bu kişilerin eylemleri, açık uygulamaların yanıltıcı kullanımlarıyla gölgelenmekte ve ilerlemeye yönelik gelişmeleri kolaylaştırmak amacıyla tasarlanan eylemler, net bir kalite kontrolü olmaksızın sadece ‘kutucuk işaretleme’ye indirgenmektedir.
Participatory Research	Participatory research refers to incorporating the views of people from relevant communities in the entire research process to achieve shared goals between researchers and the communities. This approach takes a collaborative stance that seeks to reduce the power imbalance between the researcher and those researched through a “systematic cocreation of new knowledge” (Andersson, 2018).	**Katılımlı Araştırma**	Katılımlı araştırma, araştırmacılar ve topluluklar arasında ortak hedeflere ulaşmak amacıyla, ilgili topluluklardan kişilerin görüşlerinin tüm araştırma sürecine dahil edilmesini ifade eder. Bu yaklaşım, ‘yeni bilginin sistematik ortak üretimi’ yoluyla, araştırmacı ile araştırılanlar arasındaki güç dengesizliğini azaltmayı hedefleyen iş birlikçi bir duruş benimser (Andersson, 2018).

Posterior distribution	A way to summarize one's updated knowledge in Bayesian inference, balancing prior knowledge with observed data. In statistical terms, posterior distributions are proportional to the product of the likelihood function and the prior. A posterior probability distribution captures (un)certainty about a given parameter value.	**Sonsal dağılım**	Bayesyen çıkarımda, önceden sahip olunan bilgi ile gözlemlenen veriyi dengeleyerek güncellenmiş bilgiyi özetlemenin bir yoludur. İstatistiksel terimlerle, sonsal dağılımlar, olasılık fonksiyonu ile önsel bilginin çarpımına orantılıdır. Sonsal olasılık dağılımı, belirli bir parametre değeri hakkındaki belirsizliği veya kesinliği yansıtır.
HARKing	A questionable research practice termed 'Hypothesizing After the Results are Known' (HARKing). "HARKing is defined as presenting a post hoc hypothesis (i.e., one based on or informed by one's results) in a research report as if it was, in fact, a priori" (Kerr, 1998, p. 196). For example, performing subgroup analyses, finding an effect in one subgroup, and writing the introduction with a	**HARKing**	'Sonuçlar Bilindikten Sonra Hipotez Belirleme' (HARKing; Hypothesizing After the Results are Known) olarak adlandırılan şüpheli bir araştırma pratiğidir. "HARKing, sonuçlara dayanarak veya sonuçlardan etkilenerek oluşturulan bir hipotezin, araştırma raporunda sanki araştırmadan önce (a priori) belirlenmiş gibi sunulmasıdır" (Kerr, 1998, s. 196). Örneğin, alt grup analizleri yapmak, bir alt

	‘hypothesis’ that matches these results.		grupta bir etki bulmak ve ardından giriş bölümünü bu sonuçlarla uyumlu bir 'hipotez' oluşturacak şekilde yazmak buna örnek olarak gösterilebilir.
Matthew effect (in science)	Named for the ‘rich get richer; poor get poorer’ paraphrase of the Gospel of Matthew. Eminent scientists and early-career researchers with a prestigious fellowship are disproportionately attributed greater levels of credit and funding for their contributions to science while relatively unknown or early-career researchers without a prestigious fellowship tend to get disproportionately little credit for comparable contributions. The impact is a substantial cumulative advantage that results from modest	**Matthew Etkisi (Bilimde)**	Matta İncili’nden alınan “zengin daha da zenginleşir; yoksul daha da yoksullaşır” ifadesinin bir uyarlamasından adını alır. Saygın bir bursu olan seçkin bilim insanları ve erken kariyer araştırmacıları, bilime yaptıkları katkılar için orantısız bir şekilde daha fazla kredi ve fona sahip olurken, saygın bir bursu olmayan nispeten bilinmeyen veya kariyerinin başındaki araştırmacılar, karşılaştırılabilir katkılar için orantısız bir şekilde az kredi alma eğilimindedir. Bu durum, mütevazı başlangıç karşılaştırmalı avantajlarından (ya da dezavantajlarından)

	initial comparative advantages (and vice versa).		kaynaklanan önemli bir kümülatif avantajdır.